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### Information processing apparatus, information processing method, and information processing system

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#### Abstract

An information processing apparatus including a display control unit that causes an image captured by any of one or a plurality of imaging devices to be displayed in a captured image display region that is a part of a display unit, causes second information related to the one or plurality of imaging devices to be displayed in a captured image hiding region that is a part of the display unit and different from the captured image display region, and causes first information related to the one or plurality of corresponding imaging devices to be displayed in the captured image display region in accordance with the second information being displayed and the one or plurality of imaging devices being in a predetermined state.

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## Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a continuous application of U.S. patent application Ser. No. 18/297,939, filed on Apr. 10, 2023, which is a continuation of U.S. patent application Ser. No. 17/676,480, filed on Feb. 21, 2022, now U.S. Pat. No. 11,683,580, which is a continuation application of U.S. patent application Ser. No. 15/781,947, filed on Jun. 6, 2018, now U.S. Pat. No. 11,277,568, which is a U.S. National Phase of

International Patent Application No. PCT/JP2016/078217 filed on Sep. 26, 2016, which claims the benefit of priority from Japanese Patent Application No. JP 2015-256141 filed in the Japan Patent Office on Dec. 28, 2015. Each of the above-referenced applications is hereby incorporated herein by reference in its entirety.

## TECHNICAL FIELD

(1) The present disclosure relates to an information processing apparatus, an information processing method, a program, and an information processing system.

## BACKGROUND ART

(2) Imaging devices such as digital still cameras and digital video cameras (hereinafter collectively referred to as a “digital camera” in some cases) have been widely spread, and their types have been diversified. For example, Patent Literature 1 discloses an example of an imaging device configured as a digital camera.

(3) In addition, in recent years, an imaging device has been proposed that can be manipulated from an external device such as a smartphone or a remote controller by utilizing a wireless network based on a standard such as Wi-Fi (Wireless Fidelity) (registered trademark) or Bluetooth (registered trademark).

## CITATION LIST

Patent Literature

(4) Patent Literature 1: JP 2012-44566A

## DISCLOSURE OF INVENTION

### Technical Problem

(5) In addition, in recent years, the usage form and usage scenario of an imaging device have been diversified further along with achievement of size reduction of an image sensor and an optical system. In particular, in the case of capturing an image using an imaging device such as a digital camera, a usage form in which a user captures an image while checking information or the like that indicates various states such as a zoom magnification and a remaining battery level, in addition to a so-called preview image, may be expected. Expecting such circumstances, provision of an easier-to-use user interface (UI) that can present various types of information utilizing a restricted display region more effectively is demanded.

(6) Therefore, the present disclosure proposes an information processing apparatus, an information processing method, a program, and an information processing system that can present various types of information related to image capturing in a more suitable manner.

### Solution to Problem

(7) According to the present disclosure, there is provided an information processing apparatus including: a display control unit configured to cause an image captured by any of one or a plurality of imaging devices to be displayed in a captured image display region that is a part of a display unit, cause second information related to the one or plurality of imaging devices to be displayed in a captured image hiding region that is a part of the display unit and different from the captured image display region, and cause first information related to the one or plurality of corresponding imaging devices to be displayed in the captured image display region in accordance with the second information being displayed and the one or plurality of imaging devices being in a predetermined state.

(8) In addition, according to the present disclosure, there is provided an information processing method including, by a processor: causing an image captured by any of one or a plurality of imaging devices to be displayed in a captured image display region that is a part of a display unit; causing second information related to the one or plurality of imaging devices to be displayed in a captured image hiding region that is a part of the display unit and different from the captured image display region; and causing first information related to the one or plurality of corresponding imaging devices to be displayed in the captured image display region in accordance with the

second information being displayed and the one or plurality of imaging devices being in a predetermined state.

(9) In addition, according to the present disclosure, there is provided a program causing a computer to execute: causing an image captured by any of one or a plurality of imaging devices to be displayed in a captured image display region that is a part of a display unit; causing second information related to the one or plurality of imaging devices to be displayed in a captured image hiding region that is a part of the display unit and different from the captured image display region; and causing first information related to the one or plurality of corresponding imaging devices to be displayed in the captured image display region in accordance with the second information being displayed and the one or plurality of imaging devices being in a predetermined state.

(10) In addition, according to the present disclosure, there is provided an information processing system including: one or a plurality of imaging devices configured to capture an image; and a display control device including a communication unit configured to communicate with the one or plurality of imaging devices and a display control unit configured to cause the image captured by any of the one or plurality of imaging devices to be displayed in a captured image display region that is a part of a display unit. The display control unit causes second information related to the one or plurality of imaging devices to be displayed in a captured image hiding region that is a part of the display unit and different from the captured image display region, and causes first information related to the one or plurality of corresponding imaging devices to be displayed in the captured image display region in accordance with the second information being displayed and the one or plurality of imaging devices being in a predetermined state.

#### Advantageous Effects of Invention

(11) According to the present disclosure as described above, an information processing apparatus, an information processing method, a program, and an information processing system that can present various types of information related to image capturing in a more suitable manner are provided.

(12) Note that the effects described above are not necessarily limitative. With or in the place of the above effects, there may be achieved any one of the effects described in this specification or other effects that may be grasped from this specification.

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## Description

### BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 is an explanatory diagram for describing an example of a schematic system configuration of an information processing system according to a first embodiment of the present disclosure.

(2) FIG. 2 is a front view of a control device according to the embodiment.

(3) FIG. 3 is a rear view of the control device according to the embodiment.

(4) FIG. 4 is a right side view of the control device according to the embodiment.

(5) FIG. 5 is a left side view of the control device according to the embodiment.

(6) FIG. 6 is a bottom view of the control device according to the embodiment.

(7) FIG. 7 is a front view of an imaging device according to the embodiment.

(8) FIG. 8 is a rear view of the imaging device according to the embodiment.

(9) FIG. 9 is a right side view of the imaging device according to the embodiment.

(10) FIG. 10 is a left side view of the imaging device according to the embodiment.

(11) FIG. 11 is a top view of the imaging device according to the embodiment.

(12) FIG. 12 is a bottom view of the imaging device according to the embodiment.

(13) FIG. 13 is a block diagram showing an example of a functional configuration of the information processing system according to the embodiment.

(14) FIG. 14 is an explanatory diagram for describing an example of a UI according to a comparative example.

(15) FIG. 15 is an explanatory diagram for describing an example of a UI of the control device according to the embodiment.

(16) FIG. 16 is an explanatory diagram for describing an example of a UI of the control device according to the embodiment.

(17) FIG. 17 is an explanatory diagram for describing an example of a UI of the control device according to the embodiment.

(18) FIG. 18 is an explanatory diagram for describing an example of a UI of the control device according to the embodiment.

(19) FIG. 19 is an explanatory diagram for describing an example of display information to be displayed on the UI according to the embodiment.

(20) FIG. 20 is an explanatory diagram for describing an example of display information to be displayed on the UI according to the embodiment.

(21) FIG. 21 is an explanatory diagram for describing an example of display information to be displayed on the UI according to the embodiment.

(22) FIG. 22 is an explanatory diagram for describing an example of display information to be displayed on the UI according to the embodiment.

(23) FIG. 23 is an explanatory diagram for describing an example of display information to be displayed on the UI according to the embodiment.

(24) FIG. 24 is an explanatory diagram for describing an example of a UI of the imaging device according to the embodiment.

(25) FIG. 25 is an explanatory diagram for describing an example of a manipulation menu of the control device according to the embodiment.

(26) FIG. 26 is an explanatory diagram for describing an example of a manipulation menu of the imaging device according to the embodiment.

(27) FIG. 27 is an explanatory diagram for describing an example of a schematic system configuration of an information processing system according to a second embodiment of the present disclosure.

(28) FIG. 28 is an explanatory diagram for describing an example of a UI of a control device according to the embodiment.

(29) FIG. 29 is a block diagram showing a hardware configuration example of a control device according to one embodiment of the present disclosure.

## MODE(S) FOR CARRYING OUT THE INVENTION

(30) Hereinafter, (a) preferred embodiment(s) of the present disclosure will be described in detail with reference to the appended drawings. Note that, in this specification and the appended drawings, structural elements that have substantially the same function and structure are denoted with the same reference numerals, and repeated explanation of these structural elements is omitted.

(31) Note that description will be provided in the following order. 1. First Embodiment 1.1. System configuration 1.2. Appearance example of apparatus 1.2.1. Configuration of control device 1.2.2. Configuration of imaging device 1.3. Functional configuration 1.4. Technical characteristics 1.4.1. Comparative example 1.4.2. UI of control device 1.4.3. Specific example of display information 1.4.4. UI of imaging device 1.4.5. Configuration of manipulation menu 1.5. Evaluation 2. Second Embodiment 2.1. System configuration 2.2. Technical characteristics 2.3. Evaluation 3. Hardware configuration 4. Conclusion

### 1. First Embodiment

#### (32) <1.1. System Configuration>

(33) First, an example of a system configuration of an information processing system according to a first embodiment of the present disclosure will be described with reference to FIG. 1. FIG. 1 is an explanatory diagram for describing an example of a schematic system configuration of the

information processing system according to the present embodiment.

(34) As shown in FIG. 1, an information processing system **1** according to the present embodiment includes a control device **10** and an imaging device **20**. The imaging device **20** is equivalent to a device for capturing an image such as a still image or a moving image, such as a so-called digital camera. In addition, the control device **10** is equivalent to a device for controlling the operation of the imaging device **20** from the outside, such as a so-called remote controller or the like. Note that the control device **10** may be a user terminal such as a so-called smartphone or the like.

(35) The control device **10** and the imaging device **20** are connected to be capable of communicating with each other via a wireless network **N1**. A specific example of the network **N1** includes a network based on the Wi-Fi (registered trademark) standard. In the case of applying the network **N1** based on the Wi-Fi standard, for example, it is possible to achieve a communication form as shown in FIG. 1 by causing the control device **10** to operate as an access point and connecting the imaging device **20** as a station to the control device **10**.

(36) In addition, the control device **10** according to the present embodiment is configured to be capable of controlling the operation (for example, the operation related to image capturing) of the imaging device **20** connected via the network **N1**. That is, the imaging device **20** is capable of capturing an image such as a moving image or a still image on the basis of an instruction transmitted from the control device **10** via the network **N1**.

(37) More specifically, the control device **10** includes a display unit **111** and an input unit such as a recording button **121** and an up-down button **122** which will be described later or the like. The display unit **111** includes a liquid crystal display or the like, and presents various types of information as display information. In addition, the input unit such as the recording button **121** and the up-down button **122** or the like is configured as a manipulation device such as a so-called button, switch, or the like, and accepts an input from a user.

(38) On the basis of such a configuration, the control device **10** presents an image (for example, a preview image) captured by the imaging device **20** and transmitted from the imaging device **20** via the network **N1**, for example, to the user via the display unit **111**. In addition, the control device **10** may acquire information indicating the setting and state of the imaging device **20** from the imaging device **20** via the network **N1**, and may present the acquired information to the user via the display unit **111**.

(39) In addition, the control device **10** is also capable of instructing the imaging device **20** to perform the operation related to image capturing via the network **N1** on the basis of an input from the user via the input unit such as the recording button **121** and the up-down button **122** or the like. Accordingly, it is also possible for the user to instruct the imaging device **20** to perform the operation related to image capturing via the control device **10** while checking a preview image displayed on the display unit **111** of the control device **10**, for example.

(40) Note that the operation of the imaging device **20** that can be controlled by the control device **10** via the network **N1** is not limited only to the operation of capturing an image such as a moving image or a still image. For example, the control device **10** is also capable of controlling an operation related to the setting for capturing an image, such as an adjustment of zoom or shutter speed or the like, via the network **N1**. Note that, in the case of describing “the operation related to image capturing” in the following description, it is assumed that an operation related to the setting for capturing an image may also be included in addition to the operation of capturing an image. In addition, the above-described function may be achieved by, for example, installing an application created utilizing an application programming interface (API) for controlling the operation of the imaging device **20** via a network in the control device **10**. As a matter of course, an application for achieving the above-described function may be incorporated in the control device **10** in advance.

(41) Note that the control device **10** and the imaging device **20** may be connected to be capable of communicating with each other via another network different from the network **N1**, in addition to the network **N1**. As a specific example, the control device **10** and the imaging device **20** may be

connected via a network based on the Bluetooth (registered trademark) standard. In this case, for example, the control device **10** and the imaging device **20** may selectively switch between the above-described network based on the Wi-Fi standard and a network based on the Bluetooth standard.

(42) More specifically, communication based on the Bluetooth standard is restricted in communication band as compared with communication based on the Wi-Fi standard, but is advantageous in that the operation can be performed with less consumption power. Thus, in a situation where an image is not transmitted/received between the control device **10** and the imaging device **20**, for example, communication based on the Wi-Fi standard may be stopped temporarily, and only control information may be transmitted/received through communication based on the Bluetooth standard. Examples of the situation where an image is not transmitted/received between the control device **10** and the imaging device **20** include a situation where the control device **10** is brought into the so-called sleep state, and the like.

(43) Note that the imaging device **20** may be configured such that, in the case where the imaging device **20** is not connected to the control device **10** via the network **N1**, the imaging device **20** itself can operate independently. Thus, the imaging device **20** may be provided with a manipulation interface (for example, an input interface such as a button, an output interface such as a display, and the like) for manipulating the imaging device **20**.

(44) An example of a schematic system configuration of the information processing system according to the first embodiment of the present disclosure has been described above with reference to FIG. 1.

(45) <1.2. Appearance Example of Apparatus>

(46) Next, an example of schematic configurations of the control device **10** and the imaging device **20** according to the first embodiment of the present disclosure will be described.

(47) {1.2.1. Configuration of Control Device}

(48) First, an example of a schematic configuration of the control device **10** according to the present embodiment will be described with reference to FIG. 2 to FIG. 6. FIG. 2 to FIG. 6 are five side views showing an example of a schematic configuration of the control device **10** according to the present embodiment, and respectively show the front view, the rear view, the right side view, the left side view, and the bottom view of the control device **10**. Note that the present description will be made assuming the direction corresponding to the top to the bottom of the front view shown in FIG. 2 as the vertical direction of the control device **10**, and assuming the direction corresponding to the left to the right as the lateral direction of the control device **10**.

(49) The control device **10** has a plate-like housing **11** whose front surface and rear surface have been formed in a substantially rectangular shape as shown in FIG. 2 and FIG. 3. In addition, the top surface and the bottom surface of the housing **11** have such a curved shape that arcs are formed from the front surface side to the rear surface side as shown in FIG. 4 and FIG. 5.

(50) The display unit **111** which is substantially rectangular is provided on the front surface of the housing **11** as shown in FIG. 2, and the recording button **121** is provided under the display unit **111**. The display unit **111** is equivalent to the display unit **111** described with reference to FIG. 1. In addition, the recording button **121** is an input interface for instructing the imaging device **20** connected to the control device **10** via a network to start and stop recording of a moving image or to capture a still image. Note that another function may be assigned to the recording button **121**. In addition, it may be configured such that functions assigned to the recording button **121** can be switched selectively.

(51) On the right side surface of the housing **11**, the up-down button **122** formed in a long shape along the vertical direction of the housing **11** is provided as shown in FIG. 4. The up-down button **122** is an input interface for performing a manipulation such as selection from a menu for manipulating the control device **10** or switching between modes or functions. The up-down button **122** is configured such that the upper end side and the lower end side can each be pressed down



with the center in the vertical direction serving as the base point. As a specific example, the up-down button **122** may include a so-called seesaw switch, and may be configured such that the upper end side and the lower end side can be pressed down exclusively.

(52) On the left side surface of the housing **11**, a power button **123** and a menu button **124** are provided as shown in FIG. 5. For example, in the example shown in FIG. 5, the power button **123** and the menu button **124** are provided on the left side surface of the housing **11** so as to align in the vertical direction of the housing **11**. The power button **123** is an input interface for turning on and off the power of the control device **10**. In addition, the menu button **124** is an input interface for causing the manipulation menu of the control device **10** to be displayed on the display unit **111**. Note that another function may be assigned to the menu button **124**. In addition, it may be configured such that functions assigned to the menu button **124** can be switched selectively.

(53) In addition, as shown in FIG. 2 and FIG. 6, on the rear surface of the housing **11**, a projecting portion **191** extending toward the vertical direction of the housing **11** such that side surfaces are formed in the lateral direction is provided. In addition, a connector **192** is provided on one of the lateral ends of the projecting portion **191**, as shown in FIG. 3. The connector **192** is a component for electrically connecting a battery charger or another external unit to the control device **10**. In the example shown in FIG. 3, for example, the connector **192** is provided on an end of the projecting portion **191** that is equivalent to the left side surface side of the housing **11**.

(54) In addition, a fitting portion **193** is provided on each of the left and right side surfaces of the projecting portion **191** as shown in FIG. 3 to FIG. 5. For example, the fitting portion **193** is formed as a long shaped recess along the direction in which the projecting portion **191** extends (that is, the vertical direction of the housing **11**), as shown in FIG. 4 and FIG. 5. In this manner, by providing the projecting portion **191** on the rear surface of the housing **11** and forming the fitting portions **193** in the side surfaces of the projecting portion **191**, it is possible to attach a supporting member, such as a harness or a strap for mounting the control device **10** on an area (for example, an arm or the like) of the user, to the rear surface of the housing **11**. As a more specific example, the supporting member may be provided with fitting portions (for example, projecting portions) formed to be fittable into the left and right fitting portions **193**, respectively. With such a configuration, by fitting the fitting portions provided on the supporting member side into the fitting portions **193** on the control device **10** side and sandwiching the projecting portion **191** on the control device **10** side by the fitting portions on the supporting member side from the lateral direction, it is possible to mount the supporting member on the rear surface of the control device **10**.

(55) Note that the configuration of the projecting portion **191** and the fitting portions **193** is a mere example, and the configuration of portions equivalent to the projecting portion **191** and the fitting portions **193** is not particularly limited as long as it is possible to mount a supporting member such as a harness or a strap on the rear surface of the housing **11** of the control device **10**.

(56) An example of a schematic configuration of the control device **10** according to the present embodiment has been described above with reference to FIG. 2 to FIG. 6.

(57) {1.2.2. Configuration of Imaging Device}

(58) Next, an example of a schematic configuration of the imaging device **20** according to the present embodiment will be described with reference to FIG. 7 to FIG. 12. FIG. 7 to FIG. 12 are six side views showing an example of a schematic configuration of the imaging device **20** according to the present embodiment, and respectively show the front view, the rear view, the right side view, the left side view, the top view, and the bottom view of the imaging device **20**. Note that the present description will be made assuming the direction corresponding to the top to the bottom of the front view shown in FIG. 7 as the vertical direction of the imaging device **20**, and assuming the direction corresponding to the left to the right as the lateral direction of the imaging device **20**. In addition, in the lateral direction of the right side view shown in FIG. 9, the left direction (that is, the front view side) is assumed as the front of the imaging device **20**, and the right direction (that is, the rear surface side) is assumed as the back of the imaging device **20**.

(59) As shown in FIG. 7 to FIG. 12, the imaging device **20** includes a substantially cuboid housing **21** whose right side surface and left side surface have been formed in a substantially rectangular shape, and a substantially cylindrical lens unit **22** provided on the upper side of the front surface of the housing **21** so as to protrude forward. The top surface of the housing **21** is formed as such a curved shape that an arc is formed from the right side surface to the left side surface. In addition, an axis L of the substantially cylindrical lens unit **22** substantially matches the direction from the back to the front of the imaging device **20**. In addition, the lens unit **22** is formed such that a part of a surface positioned on the upper side of the peripheral surface of the cylinder continues to the curved top surface of the housing **21**. In addition, the horizontal width (that is, the width between the right side surface and the left side surface) of the housing **21** and the diameter of the lens unit **22** substantially match. Note that the lens unit **22** may be formed integrally with the housing **21**.

(60) In addition, a lens optical system **261** is provided on the front side of the lens unit **22**. At this time, the optical axis direction of the lens optical system **261** substantially matches the axis L of the substantially cylindrical lens unit **22**.

(61) In addition, on the front surface of the housing **21**, a sound collection unit **270** is provided under the lens unit **22**.

(62) On the top surface of the housing **21**, a recording button **221** and a power button **224** are provided as shown in FIG. 11. The recording button **221** is an input interface for instructing the imaging device **20** to start and stop recording of a moving image or to capture a still image. Note that another function may be assigned to the recording button **221**. The recording button **221** is provided such that the top surface of the button protrudes from the top surface of the housing **21** as shown in FIG. 9 and FIG. 10. In addition, the power button **224** is provided such that the top surface of the button is buried in the top surface of the housing **21**.

(63) On the left side surface of the housing **21**, a display unit **211**, an upper button **222**, a lower button **223**, and a menu button **225** are provided as shown in FIG. 10. The display unit **211** includes a liquid crystal display or the like, and presents various types of information as display information. The upper button **222** and the lower button **223** are input interfaces for performing a manipulation such as selection from a menu for manipulating the imaging device **20** or switching between modes or functions. In addition, the menu button **225** is an input interface for causing the manipulation menu of the imaging device **20** to be displayed on the display unit **211**. Note that another function may be assigned to the menu button **225**.

(64) The upper button **222** and the lower button **223** are provided to align in the vertical direction close to the display unit **211** so as to be positioned on the right side with respect to the display unit **211** (that is, the front side of the imaging device **20**). Note that the upper button **222** and the lower button **223** are provided such that the upper button **222** is positioned on the upper side, and the lower button **223** is positioned on the lower side. That is, the positional relationship of the display unit **211** with the upper button **222** and the lower button **223** is similar to the positional relationship between the display unit **111** and the up-down button **122** in the control device **10** described with reference to FIG. 2 and FIG. 4. Thus, it is possible for the user to intuitively manipulate the upper button **222** and the lower button **223** with a similar sense to the case of manipulating the up-down button **122** in the control device **10**.

(65) In addition, the menu button **225** is provided close to the display unit **211** so as to be positioned on the left side with respect to the display unit **211** (that is, the back side of the imaging device **20**).

(66) On the right side surface the housing **21**, a battery compartment is provided as shown in FIG. 9, and a part of the right side surface of the housing **21** is configured as a battery cover **293** that covers the battery compartment. The battery cover **293** is configured to be attachable/detachable to/from the housing **21** by being slid backward. That is, by removing the battery cover **293**, a battery holder is exposed, and it is possible to insert/remove the battery into/from the battery holder. Note that an end of the battery cover **293** on the front side may be supported pivotably on

the housing **21**. On the basis of such a configuration, the battery cover **293** may be configured to be openable/closable by pivoting the end on the back side with the end on the front side supported by the housing **21** serving as the base point.

(67) In addition, at least a part of the end of the battery cover **293** on the back side is extended to the rear surface side of the housing **21**, and a lock switch **294** is provided on the portion extended to the rear surface side as shown in FIG. **8**. The lock switch **294** is configured as a so-called slide switch. By being slid upward in a state where the battery cover **293** is mounted on the housing **21**, the lock switch **294** restricts backward sliding of the battery cover **293** (that is, an action for removing the battery cover **293**).

(68) On the bottom surface of the housing **21**, a fitting portion **291** and a memory card slot **292** are provided as shown in FIG. **12**. The memory card slot **292** is configured such that a memory card which is a semiconductor recording medium can be attached/detached. With a memory card inserted into the memory card slot **292**, it is possible for the imaging device **20** to record or read out image data such as a moving image or a still image.

(69) The fitting portion **291** is a component for mounting a supporting member for supporting the imaging device **20**, such as a tripod, a harness, a strap, or the like, on the bottom surface of the imaging device **20**. For example, the fitting portion **291** may be formed as a so-called screw hole (female screw) obtained by forming a thread on the inner surface of a recess. In this case, by screwing a screw provided on the supporting member side into the fitting portion **291**, it is possible to mount the supporting member on the bottom surface of the imaging device **20**.

(70) Note that the above-described configuration of the fitting portion **291** is a mere example, and the configuration of the fitting portion **291** is not particularly limited as long as a supporting member such as a tripod, a harness, a strap, or the like can be mounted on the bottom surface of the imaging device **20**.

(71) An example of a schematic configuration of the imaging device **20** according to the present embodiment has been described above with reference to FIG. **7** to FIG. **12**.

(72) <1.3. Functional Configuration>

(73) Subsequently, an example of a functional configuration of the information processing system **1** according to the present embodiment will be described with reference to FIG. **13**, particularly paying attention to a functional configuration of each of the control device **10** and the imaging device **20**. FIG. **4** is a block diagram showing an example of the functional configuration of the information processing system **1** according to the present embodiment.

(74) (Imaging Device **20**)

(75) First, description will be made paying attention to the functional configuration of the imaging device **20** according to the present embodiment. As shown in FIG. **13**, the imaging device **20** includes an imaging unit **260**, an output unit **210**, an input unit **220**, a communication unit **230**, a control unit **240**, and a storage unit **250**. Note that the imaging device **20** may include the sound collection unit **270**. In addition, the control unit **240** includes an imaging control unit **241**, an input analysis unit **243**, and an output control unit **245**.

(76) The communication unit **230** is a communication interface for each component in the imaging device **20** to communicate with an external device (for example, the control device **10**) via a network. Examples of the network include the wireless network **N1** described with reference to FIG. **1**. As a matter of course, as long as it is a network that can connect the imaging device **20** and an external device, its form is not particularly limited. Note that, in the following description, in the case where each component in the imaging device **20** transmits/receives information to/from an external device via a network, it is assumed that the information is transmitted/received via the communication unit **230** even in the case where there is no particular description.

(77) The imaging unit **260** includes an image sensor such as a complementary metal-oxide semiconductor (CMOS) image sensor or a charge coupled device (CCD) image sensor, and an optical system such as a lens, and is a component for capturing an image of a subject. For example,

the imaging unit **260** may be configured such that a focal position and a so-called optical zoom magnification can be adjusted by shifting the optical system such as a lens along the optical axis direction. Note that the operation of the imaging unit **260** is controlled by the control unit **240**. In addition, the imaging unit **260** outputs data (hereinafter referred to as “image data” as well) of a captured image to the control unit **240**.

(78) The sound collection unit **270** includes a sound collection device such as a so-called microphone or the like, and collects external sound (for example, ambient sound, voice, or the like). The operation of the sound collection unit **270** is controlled by the control unit **240**. In addition, the sound collection unit **270** outputs data indicating the result of sound collection (hereinafter referred to as “acoustic data” as well) to the control unit **240**.

(79) The storage unit **250** is a storage area for storing various types of data temporarily or permanently. For example, image data based on the result of imaging of the imaging unit **260** and acoustic data based on the result of sound collection performed by the sound collection unit **270** are recorded in the storage unit **250**. In addition, data for the imaging device **20** to execute various functions may be recorded in the storage unit **250**. As a specific example, data (for example, a library) for executing various applications, management data for managing various settings and the like, and the like may be recorded in the storage unit **250**.

(80) The input unit **220** is an input interface for the user to input various types of information to the imaging device **20**. The input unit **220** may include an input device for the user to input information, such as a button, a switch, a touch panel, or the like, for example. For example, each of the recording button **221**, the power button **224**, the upper button **222**, the lower button **223**, and the menu button **225** described with reference to FIG. 7 to FIG. 12 is equivalent to an example of the input unit **220**. Then, the input unit **220** outputs control information indicating manipulation contents input by the user to the control unit **240**.

(81) The output unit **210** is an output interface for the imaging device **20** to present various types of information to the user. The output unit **210** may include a display device that outputs an image such as a still image or a moving image, such as a so-called display or the like, for example. For example, the display unit **211** described with reference to FIG. 7 to FIG. 12 is equivalent to an example of the output unit **210**. In addition, the output unit **210** may include an acoustic device that outputs sound, such as a speaker or the like, for example. In addition, the output unit **210** may include a vibration device, such as a so-called vibrator or the like, that, by vibrating in a pattern corresponding to information to be a presentation target, presents the information to the user.

(82) The input analysis unit **243** acquires control information indicating a user input from the input unit **220**, and recognizes instruction contents from the user on the basis of the acquired control information. For example, on the basis of control information indicating manipulation contents via a button, the input analysis unit **243** may specify a function associated with the button (that is, a function instructed by the user to be executed). In addition, on the basis of control information indicating manipulation contents via a button, a touch panel, or the like, the input analysis unit **243** may specify information selected by the user as a manipulation target among various types of information presented via the output unit **210**. In the foregoing manner, the input analysis unit **243** recognizes the instruction contents from the user on the basis of control information acquired from the input unit **220**, and outputs information indicating the instruction contents to the imaging control unit **241** or the output control unit **245** in accordance with the instruction contents.

(83) In addition, the input analysis unit **243** may acquire information indicating the instruction contents from the user input via the control device **10** from the control device **10** via the network. In this case, the input analysis unit **243** outputs the acquired information indicating the instruction contents to the imaging control unit **241** or the output control unit **245** in accordance with the instruction contents.

(84) The imaging control unit **241** controls the operation related to image capturing. For example, the imaging control unit **241** acquires information indicating an instruction of the operation related

to image capturing based on an input from the user from the input analysis unit **243**, and controls the operation of the imaging unit **260** on the basis of the acquired information. In addition, the imaging control unit **241** acquires image data from the imaging unit **260** as the result of imaging, and outputs the acquired image data to the output control unit **245**.

(85) In addition, the imaging control unit **241** may control the operation of the sound collection unit **270**. For example, the imaging control unit **241** acquires information indicating an instruction of the operation related to sound collection based on an input from the user from the input analysis unit **243**, and controls the operation of the sound collection unit **270** on the basis of the acquired information. In addition, the imaging control unit **241** acquires acoustic data from the sound collection unit **270** as the result of sound collection, and outputs the acquired acoustic data to the output control unit **245**.

(86) In addition, the imaging control unit **241** may update various settings and operation modes of each of the imaging unit **260** and the sound collection unit **270** on the basis of an instruction from the user acquired via the input analysis unit **243**.

(87) In addition, the imaging control unit **241** may monitor the state of the imaging unit **260** or the sound collection unit **270**, and may output information indicating the result of monitoring to the output control unit **245**. As a specific example, the imaging control unit **241** may monitor the temperature of the imaging unit **260** on the basis of information indicating the result of sensing of the temperature of the imaging unit **260** output from a predetermined temperature sensor (illustration omitted), and may output information indicating the result of monitoring to the output control unit **245**. In addition, in the case where various settings and operation modes of each of the imaging unit **260** and the sound collection unit **270** are updated, the imaging control unit **241** may output information indicating the result of update to the output control unit **245**.

(88) The output control unit **245** is a component for controlling output of various types of information. For example, the output control unit **245** may acquire image data based on the result of imaging performed by the imaging unit **260** from the imaging control unit **241**, and may cause the acquired image data to be output to the output unit **210** (for example, a display or the like). In addition, the output control unit **245** may acquire acoustic data based on the result of sound collection performed by the sound collection unit **270** from the imaging control unit **241**, and may cause the acquired acoustic data to be output to the output unit **210** (for example, a microphone or the like). In addition, the output control unit **245** may cause the image data or acoustic data acquired from the imaging control unit **241** to be recorded in the storage unit **250**.

(89) In addition, the output control unit **245** may acquire information indicating the result of update of various settings and operation modes of each of the imaging unit **260** and the sound collection unit **270** from the imaging control unit **241**, and may update management data for managing the settings and operation modes on the basis of the acquired information. Note that the management data may be managed by being recorded in the storage unit **250**, for example. In addition, the output control unit **245** may cause information indicating various settings and operation modes of each of the imaging unit **260** and the sound collection unit **270** to be output to the output unit **210** (for example, a display or the like) on the basis of the management data.

(90) In addition, the output control unit **245** may acquire information indicating the result of monitoring the state of the imaging unit **260** or the sound collection unit **270** from the imaging control unit **241**. As a specific example, the output control unit **245** may acquire information indicating the result of monitoring the temperature of the imaging unit **260** from the imaging control unit **241**. Accordingly, in the case where the temperature of the imaging unit **260** indicates a value outside a predetermined range, for example, it is also possible for the output control unit **245** to sense this state as a temperature anomaly and to cause information indicating a warning of the temperature anomaly to be output to the output unit **210**.

(91) In addition, the output control unit **245** may acquire information indicating an instruction of an operation related to output of various types of information based on an input from the user from the

input analysis unit **243**, and may cause the output unit **210** to output information instructed to be output on the basis of the acquired information. As a specific example, the output control unit **245** may cause a manipulation menu for the user to instruct the imaging device **20** to execute various functions to be output to the output unit **210** (for example, a display or the like).

(92) Note that an example of a UI (for example, a manipulation screen) for the output control unit **245** to cause the above-described various types of data and information to be output to the output unit **210** will be described later in detail separately.

(93) In addition, the output control unit **245** may transmit acquired various types of information to the control device **10** via the network. As a specific example, the output control unit **245** may transmit image data (for example, a preview image or the like) based on the result of imaging performed by the imaging unit **260** to the control device **10**. Accordingly, it is possible for the control device **10** to present the image data to the user. The same also applies to acoustic data based on the result of sound collection performed by the sound collection unit **270**. In addition, the output control unit **245** may transmit image data or acoustic data recorded in the storage unit **250** to the control device **10**. In addition, the output control unit **245** may transmit information indicating various settings and operation modes of each of the imaging unit **260** and the sound collection unit **270** to the control device **10**. Accordingly, it is possible for the control device **10** to notify the user of the settings and operation modes. Similarly, the output control unit **245** may transmit information based on the result of monitoring the state of the imaging unit **260** or the sound collection unit **270** to the control device **10**. Accordingly, it is possible for the control device **10** to notify the user of the state of the imaging unit **260** or the sound collection unit **270**.

(94) (Control Device **10**)

(95) Next, description will be made paying attention to the functional configuration of the control device **10** according to the present embodiment. As shown in FIG. **13**, the control device **10** according to the present embodiment includes the display unit **111**, an input unit **120**, a communication unit **130**, a control unit **140**, and a storage unit **150**. In addition, the control unit **140** includes an input analysis unit **141** and an output control unit **143**. Note that the control unit **140** is equivalent to an example of the “imaging control unit” in the control device **10**.

(96) The communication unit **130** is a communication interface for each component in the control device **10** to communicate with an external device (for example, the imaging device **20**) via a network. Note that, as the communication unit **130**, a component similar to the aforementioned communication unit **230** may be applied. Note that, in the following description, in the case where each component in the control device **10** transmits/receives information to/from an external device via a network, it is assumed that the information is transmitted/received via the communication unit **130** even in the case where there is no particular description.

(97) The storage unit **150** is a storage area for storing various types of data temporarily or permanently. For example, data for the control device **10** to execute various functions may be recorded in the storage unit **150**. As a specific example, data (for example, a library) for executing various applications, management data for managing various settings and the like, and the like may be recorded in the storage unit **150**.

(98) The input unit **120** is an input interface for the user to input various types of information to the control device **10**. The input unit **120** may include an input device for the user to input information, such as a button, a switch, a touch panel, or the like, for example. For example, each of the recording button **121**, the up-down button **122**, the power button **123**, and the menu button **124** described with reference to FIG. **2** to FIG. **6** is equivalent to an example of the input unit **120**. Then, the input unit **120** outputs control information indicating manipulation contents input by the user to the control unit **140**.

(99) The output unit **110** is an output interface for the control device **10** to present various types of information to the user. The output unit **110** may include a display device that outputs image information such as a still image or a moving image, such as a so-called display or the like, for

example. For example, the display unit **111** described with reference to FIG. 2 to FIG. 6 is equivalent to an example of the output unit **110**. In addition, the output unit **110** may include an acoustic device that outputs sound, such as a speaker or the like, for example. In addition, the output unit **110** may include a vibration device, such as a so-called vibrator or the like, that, by vibrating in a pattern corresponding to information to be a presentation target, presents the information to the user.

(100) The input analysis unit **141** acquires control information indicating a user input from the input unit **120**, and recognizes instruction contents from the user on the basis of the acquired control information. For example, the input analysis unit **141** may specify a function associated with the button (that is, a function instructed by the user to be executed) on the basis of control information indicating manipulation contents via a button. In addition, the input analysis unit **141** may specify information selected by the user as a manipulation target among various types of information presented via the output unit **110** on the basis of control information indicating manipulation contents via a button, a touch panel, or the like. In the foregoing manner, the input analysis unit **141** recognizes the instruction contents from the user on the basis of the control information acquired from the input unit **120**.

(101) Then, in accordance with the instruction contents from the user, the input analysis unit **141** determines an output destination of information indicating the instruction contents. For example, in the case where the instruction contents from the user indicate output of information via the output unit **110**, the input analysis unit **141** outputs the information indicating the instruction contents to the output control unit **143**. In addition, as another example, in the case where the instruction contents from the user indicate an instruction related to control of the operation of the imaging device **20**, the input analysis unit **141** transmits the information indicating the instruction contents to the imaging device **20** via the network.

(102) The output control unit **143** is a component for controlling output of various types of information. For example, the output control unit **143** may acquire information indicating an instruction of an operation related to output of various types of information based on an input from the user from the input analysis unit **141**, and on the basis of the acquired information, may cause the output unit **110** to output information instructed to be output. As a specific example, the output control unit **143** may cause the output unit **110** (for example, a display or the like) to output a manipulation menu for the user to instruct the control device **10** to execute various functions.

(103) In addition, the output control unit **143** may cause the output unit **110** to output information indicating the state or setting of the control device **10**. As a specific example, the output control unit **143** may acquire information indicating the state of communication with the imaging device **20** from the communication unit **130**, and on the basis of the acquired information, may notify the user of the state of the communication via the output unit **110**. In addition, as another example, the output control unit **143** may acquire information indicating the remaining battery level of the control device **10** using a system call or the like, for example, and on the basis of the acquired information, may notify the user of information (for example, a warning or the like) corresponding to the remaining battery level via the output unit **110**.

(104) In addition, the output control unit **143** may acquire various types of information from the imaging device **20** via the network, and may cause the output unit **110** to output the acquired information. For example, the output control unit **143** may acquire image data (for example, a preview image or the like) or acoustic data from the imaging device **20**, and may present the acquired image data or acoustic data to the user via the output unit **110**. In addition, the output control unit **143** may acquire information indicating various settings and operation modes of the imaging device **20** from the imaging device **20**, and on the basis of the acquired information, may notify the user of the settings and operation modes via the output unit **110**. Similarly, the output control unit **143** may acquire information indicating the state of the imaging device **20** (that is, information based on the result of monitoring the state) from the imaging device **20**, and on the

basis of the acquired information, may notify the user of the state of the imaging device **20** via the output unit **110**.

(105) An example of the functional configuration of the information processing system **1** according to the present embodiment has been described above with reference to FIG. **13**, particularly paying attention to the functional configuration of each of the control device **10** and the imaging device **20**.

(106) <1.4. Technical Characteristics>

(107) Subsequently, as technical characteristics of the information processing system **1** according to the present embodiment, an example of UIs presented by the control device **10** and the imaging device **20** will be particularly described.

(108) {1.4.1. Comparative Example}

(109) First, in order for easier understanding of technical characteristics of the information processing system **1** according to the present embodiment, an example of a UI for presenting an image (for example, a preview image) captured by the imaging device **20** and information indicating the setting and state of the imaging device **20** will be described as a comparative example with reference to FIG. **14**. FIG. **14** is an explanatory diagram for describing an example of the UI according to the comparative example, and shows an example of the UI of the control device **10**.

(110) On the UI according to the comparative example, in the case where a predetermined button on the control device **10** side is pressed down, the screen to be displayed on the display unit **111** is switched cyclically among screens **V90a** to **V90c**, as shown in FIG. **14**. For example, on the screen **V90a**, a preview image is presented, and various icons indicating the settings and states of each of the imaging device **20** and the control device **10** are superimposed on the preview image. In addition, on the screen **V90b**, presentation of various icons is restricted, and only the preview image is presented. In addition, on the screen **V90c**, by restricting presentation of the preview image, more detailed information than on the screen **V90a** is presented as information indicating the settings and states of each of the imaging device **20** and the control device **10**.

(111) On the other hand, on the UI according to the comparative example, it is necessary to restrict presentation of the preview image in order to present a specific piece of information among pieces of information indicating the settings and states of each of the imaging device **20** and the control device **10**, as is understood when comparing the screen **V90a** and the screen **V90c**. In other words, on the UI according to the comparative example, it is necessary to temporarily hide the preview image in order for the user to check a specific piece of information. Thus, for example, it may be difficult for the user to perform a manipulation related to image capturing while checking both of the specific piece of information and the preview image.

(112) Therefore, in view of the above-described circumstances, the present disclosure proposes a UI capable of presenting an image such as a preview image and various types of information (for example, setting, operation mode, state, and the like) related to image capturing in a more suitable manner.

(113) {1.4.2. UI of Control Device}

(114) First, an example of a UI of the control device **10** according to the present embodiment will be described with reference to FIG. **15** to FIG. **18**. FIG. **15** to FIG. **18** are explanatory diagrams for describing an example of the UI of the control device **10** according to the present embodiment.

(115) For example, FIG. **15** shows an example of a screen displayed as a UI on the display unit **111** of the control device **10**. As shown in FIG. **15**, a screen **V10** displayed on the display unit **111** includes a region **V11** for presenting an image (for example, a preview image) and regions **V13** and **V15** for presenting information corresponding to various states and settings. For example, in the example shown in FIG. **15**, the strip region **V13** with the lateral direction serving as the longitudinal direction is provided above the region **V11**. Similarly, the strip region **V15** with the lateral direction serving as the longitudinal direction is provided under the region **V11**.

(116) In the region **V15**, information indicating the settings and states related to image capturing is



displayed. For example, in the example shown in FIG. 15, pieces of display information V151 to V156 are displayed in the region V15. Specifically, the display information V151 indicates the operation mode of the imaging device 20, and in the example shown in FIG. 15, indicates a mode of capturing a moving image. In addition, the display information V152 indicates the capturing time of video. In addition, the display information V153 indicates the format of video. In addition, the display information V154 indicates the mode of video (for example, the number of frames or the like), and in the example shown in FIG. 15, indicates that it is progressive video including 30 frames per second. In addition, the display information V155 indicates the setting of image stabilization. In addition, the display information V156 indicates the setting of viewing angle.

(117) In addition, information indicating other states other than the settings and states related to image capturing is displayed in the region V13. For example, in the example shown in FIG. 15, pieces of display information V131 to V136 are displayed in the region V13. Specifically, the display information V131 indicates the state of communication with the imaging device 20 via a network based on the Wi-Fi standard. Similarly, the display information V132 indicates the state of communication with the imaging device 20 via a network based on the Bluetooth standard. In addition, the display information V133 indicates the state of a positioning function such as a global positioning system (GPS), and in the example shown in FIG. 15, indicates that the positioning result is being acquired by a GPS provided on the imaging device 20 side. In addition, the display information V133 indicates information related to the time, and in the example shown in FIG. 15, indicates the current time. Note that, as the display information V133, information related to the time other than the current time may be displayed. For example, in the case of reproducing an already captured image, information indicating the date and time when the image is captured may be displayed as the display information V133. In addition, the display information V135 indicates the remaining battery level of the imaging device 20, and the display information V136 indicates the remaining battery level of the control device 10 itself. In this manner, on the UI of the control device 10 according to the present embodiment, information on the control device 10 side and information on the imaging device 20 side may be displayed individually regarding some states and settings.

(118) Note that, in the regions V13 and V15, information set in advance as a presentation target in accordance with the operation state of the control device 10 is displayed constantly as long as the operation state is continued. Note that the “operation state” of the control device 10 in the present description indicates a so-called operation mode to be switched intentionally in accordance with a predetermined manipulation, such as a “single mode”, a “multi mode”, or the like which will be described later. That is, information displayed constantly in the regions V13 and V15 may be switched in accordance with the operation state (operation mode) of the control device 10. Note that the case of expressing “displayed constantly” in the following description does not necessarily only indicate the case where information is always displayed in the same manner, but it is assumed that, for each predetermined operation state (operation mode), the case where information corresponding to the operation state is displayed constantly may also be included. In addition, the regions V13 and V15 are equivalent to an example of a “second region”, and display information displayed in the regions V13 and V15 is equivalent to an example of “second information”. Note that, as long as being constantly displayed in a predetermined operation state (operation mode), the second information displayed in the second region (for example, the regions V13 and V15) may be controlled to be displayed constantly only in some operation states, for example, not limited to all the operation states. In this case, for example, an operation state (operation mode) in which the second information is not displayed constantly may exist. A specific example of the operation state in which the second information is not displayed constantly includes an operation state in which a menu screen such as a manipulation menu which will be described later is displayed, for example. In addition, the second region is equivalent to an example of a “captured image hiding region”. However, a so-called preview image, an actually captured image, or the like may be displayed in

the second region.

(119) A so-called preview image, an actually captured image, or the like is displayed in the region **V11**. In addition, in the region **V11**, information indicating various settings and states may be superimposed on the displayed image. For example, in the example shown in FIG. 15, pieces of display information **V111** to **V116** are superimposed on the image displayed in the region **V11**. Specifically, the display information **V111** is information for warning that the temperature of the imaging device **20** indicates a value outside a predetermined range, and is displayed in accordance with the result of monitoring the temperature of the imaging device **20**. In addition, the display information **V112** indicates that the sound collection unit **270** of the imaging device **20** is off (that is, in a state in which sound is not collected). In addition, the display information **V113** indicates that the imaging device **20** is executing processing, and that a manipulation is temporarily restricted. In addition, the display information **V114** indicates that it is in a so-called hold state in which manipulations via some input units **120** (for example, a button and the like) in the control device **10** are temporarily restricted. In addition, the display information **V115** indicates a manipulation assigned to the up-down button **122** shown in FIG. 2 and FIG. 4, and in the example shown in FIG. 15, indicates that a zoom manipulation in image capturing has been assigned to the up-down button **122**. In addition, the display information **V116** indicates a zoom magnification in image capturing.

(120) Note that various types of information superimposed on the image displayed in the region **V11** are temporarily displayed in accordance with a predetermined trigger, and are not displayed in the case where the imaging device **20** and the control device **10** itself are in a standard state. Note that the standard state shall indicate a state used as a reference of an operation, such as a state in which an anomaly such as a temperature increase is not sensed, a state in which a default setting has been applied, or the like, for example. More specifically, the display information **V111** is displayed only in the case where a temperature anomaly of the imaging device **20** is detected. In addition, the display information **V112** is not displayed in the case where the sound collection unit **270** of the imaging device **20** is on. In addition, the display information **V113** is displayed only in the case where the imaging device **20** is executing processing. In addition, the display information **V114** is displayed only in the case of the hold state. In addition, the display information **V115** is displayed temporarily just after a manipulation is performed on the up-down button **122**, and is hidden in the case where the state in which a manipulation on the up-down button **122** is not detected has continued for a predetermined period. In addition, the display information **V116** is displayed as information indicating the set magnification only in the case where the zoom magnification has been set at another value other than 1.0. In addition, the region **V11** is equivalent to an example of a “first region”, and display information superimposed on the image displayed in the region **V11** is equivalent to an example of “first information”. In addition, the first region is equivalent to an example of a “captured image display region”.

(121) Note that the contents of the “predetermined trigger” are not particularly limited as long as it starts a transition to another state different from the standard state. A specific example of the “predetermined trigger” includes an event in which another state different from the standard state occurs along with sensing, predetermined input information to be input for causing a transition to another state different from the standard state, and the like, for example. In addition, “temporary display” means that, different from constant display, display is not presented in the standard state, but display is presented in the case where a transition is made to another state, for example. In addition, a trigger to hide information temporarily displayed is not particularly limited. For example, information “temporarily” displayed along with the occurrence of a predetermined situation may be controlled to be hidden along with the end of the situation, or may be controlled to be hidden after the lapse of a certain time from the occurrence of the situation. Note that, even in the case where the aforementioned second region (for example, the regions **V13** and **V15**) is displayed constantly in a predetermined operation state (operation mode), information (for

example, a preview image or first information) may be displayed in the first region (for example, the region **V11**), or may not be displayed. In addition, at this time, the state in which information is displayed and the state in which information is not displayed may be switched dynamically in the first region. In other words, the first information (the display information **V111** to **V116** or the like) is displayed in accordance with the second information (the display information **V131** to **V136**, **V151** to **V156**, or the like) being displayed, and the imaging device **20** and the control device **10** itself being in a predetermined state (that is, another state different from the standard state). That is, the first information is displayed only under certain conditions, whereas the second information is displayed constantly in accordance with an operation state. Thus, the display frequency of the first information is lower than the display frequency of the second information.

(122) Here, with reference to FIG. **16**, an example of an operation of the UI of the control device **10** according to the present embodiment will be described particularly paying attention to an operation related to display of information. FIG. **16** is an explanatory diagram for describing an example of the operation of the UI of the control device **10** according to the present embodiment.

(123) In FIG. **16**, a screen denoted by a reference character **V10a** shows an example of a screen presented in the standard state. That is, in the standard state, information indicating various states and settings is not superimposed on an image displayed in the region **V11**, as shown as the screen **V10a**.

(124) In addition, a screen denoted by a reference character **V10b** shows an example of a screen in the case where a manipulation on the up-down button **122** has been detected. In the example shown as the screen **V10b**, display information **V115a** indicating that a manipulation for switching the mode has been assigned to the up-down button **122** is displayed. Note that, in the case where the state in which a manipulation on the up-down button **122** is not detected has continued for a predetermined period in the state where the screen **V10b** is displayed, the display information **V115a** is hidden.

(125) In addition, a screen denoted by a reference character **V10c** shows another example of a screen in the case where a manipulation on the up-down button **122** has been detected. In the example shown as the screen **V10c**, display information **V115b** indicating that a zoom manipulation in image capturing has been assigned to the up-down button **122** is displayed. In addition, since the zoom magnification has been set at 1.1 in the example shown as the screen **V10c**, the display information **V116** indicating the magnification is displayed. Note that, in the case where the zoom magnification is changed to 1.0 (default value), the display information **V116** is hidden. In addition, in the case where the state in which a manipulation on the up-down button **122** is not detected has continued for a predetermined period in the state where the screen **V10c** is displayed, the display information **V115b** is hidden.

(126) In addition, as shown in FIG. **16**, in any of the cases of the screens **V10a** to **V10c**, information correlated in advance is displayed constantly in a predetermined layout in each of the regions **V13** and **V15**.

(127) Note that, among pieces of display information displayed on the screen **V10**, display information associated in advance with an input interface such as a button provided for the control device **10** may be controlled in display position so as to be positioned close to the input interface.

(128) For example, FIG. **17** is an explanatory diagram for describing the display position of the display information **V115** indicating a manipulation assigned to the up-down button **122**. As described with reference to FIG. **2** and FIG. **4**, the up-down button **122** is provided on the right side surface of the housing **11** of the control device **10**. On the basis of such a configuration, the display information **V115** is presented in a region positioned closer to the up-down button **122** in the screen **V10**. For example, in the example shown in FIG. **17**, the display information **V115** is displayed on the right end side of the screen **V10**. Note that, at this time, information indicating the setting based on the manipulation indicated by the display information **V115**, such as the display information **V116**, may be displayed in a region positioned close to the up-down button **122** (for example, on

the right end side of the screen **V10** in FIG. **17**), similarly to the display information **V115**.

(129) In addition, display of the display information **V115** and **V116** and the manipulating direction indicated by the display information **V115** may be controlled such that the direction (manipulating direction) indicated by the display information **V115** and the manipulating direction (that is, the vertical direction) of the up-down button **122** substantially match. For example, in the example shown in FIG. **17**, the display information **V115** indicates that the zoom magnification can be manipulated by a manipulation in the vertical direction. At this time, in the case where the upper side of the up-down button **122** is pressed down, the zoom magnification is controlled to become larger, and along with the control, the display information **V116** is updated. In addition, in the case where the lower side of the up-down button **122** is pressed down, the zoom magnification is controlled to become smaller, and along with the control, the display information **V116** is updated.

(130) Note that the control device **10** may be configured such that the orientation in which the screen **V10** is displayed can be controlled in such a manner that the top and the bottom of the screen **V10** are reversed with respect to the top and the bottom of the housing **11**. For example, FIG. **18** shows an example of the case where the screen **V10** is displayed in such a manner that the top and the bottom of the screen **V10** are reversed with respect to the top and the bottom of the housing **V11**. Here, in the example shown in FIG. **18**, in the case where the user is diametrically opposed to the screen **V10**, the up-down button **122** is positioned on the left side of the screen **V10** as seen from the user. Thus, in the example shown in FIG. **18**, the display information **V115** and **V116** may be controlled in display position so as to be displayed on the left end side of the screen **V10**.

(131) An example of the UI of the control device **10** according to the present embodiment has been described above with reference to FIG. **15** to FIG. **18**. Note that the example described above is a mere example, and the example described with reference to FIG. **15** to FIG. **18** is not necessarily a limitation. For example, as long as the first region (for example, the region **V11**) and the second region (for example, the regions **V13** and **V15**) are presented separately, the configuration of the screen **V10** is not necessarily limited to the example shown in FIG. **15**. For example, only either of the regions **V13** and **V15** may be presented. In addition, a strip second region with the vertical direction serving as the longitudinal direction may be provided on either or both of the left and right of the first region. In addition, a semi-transparent second region may be presented on the first region.

(132) {1.4.3. Specific Example of Display Information}

(133) Next, with reference to FIG. **15** and FIG. **19** to FIG. **23**, a specific example of display information displayed in each of the regions **V11**, **V13**, and **V15** in the screen **V10** will be described. FIG. **19** to FIG. **23** are explanatory diagrams for describing an example of display information to be displayed on the UI according to the present embodiment.

(134) For example, FIG. **19** shows an example of the display information **V151** indicating the operation mode of the imaging device **20** to be displayed in the region **V15** in the screen **V10** shown in FIG. **15**. Examples of the operation mode of the imaging device **20** include modes such as video capturing, still image capturing, interval shooting, loop recording, live streaming, and audio, as shown in FIG. **19**. Note that the interval shooting is a mode of capturing a still image at each interval set in advance. In addition, the loop recording is a mode for capturing a moving image within the range of a capturing time determined in advance, and in the case where a moving image is captured beyond the capturing time, a moving image (frame) captured in the past is deleted. With such a configuration, during the loop recording, a moving image captured more recently is held by priority within the range of the capturing time determined in advance. In addition, the live streaming is an operation mode for encoding a captured image or collected sound (voice) in real time and distributing streams. In addition, the audio mode is an operation mode for collecting acoustic data alone.

(135) In addition, FIG. **20** shows another example of display information to be displayed in the

region **V15** in the screen **V10** shown in FIG. **15**. As described earlier, information indicating the states and settings related to image capturing is displayed in the region **V15**. Specifically, pieces of information indicating viewing angle setting, an image stabilization mode, a state indicating effectiveness and ineffectiveness of image stabilization, a video format, an interval timer, a state of medium, and the like are displayed in the region **V15** as shown in FIG. **20**.

(136) In addition, FIG. **21** shows an example of display information to be displayed in the region **V13** in the screen **V10** shown in FIG. **15**. As described earlier, information indicating another state other than the states and settings related to image capturing is displayed in the region **V13**.

Specifically, pieces of information indicating a mode of communication (for example, the single mode and the multi mode) based on the Wi-Fi standard, a state of communication based on the Bluetooth standard, a state of a positioning function such as GPS, a setting situation of an airplane mode, the remaining battery level of the control device **10**, the remaining battery level of the imaging device **20**, and the like are displayed in the region **V13** as shown in FIG. **21**.

(137) Note that the single mode among the communication modes based on the Wi-Fi standard is a mode in which the control device **10** and the imaging device **20** are connected in a one-to-one manner as shown in FIG. **1**. In addition, the multi mode is a mode in which a plurality of imaging devices **20** are connected to the control device **10**. In the multi mode, it is possible to check the state of each of the plurality of imaging devices **20** via the control device **10**, and to control the setting and operation of each of the plurality of imaging devices **20** via the control device **10**. Note that an example of the multi mode will be described later separately as a second embodiment.

(138) In addition, FIG. **22** shows an example of display information to be superimposed on an image displayed in the region **V11** in the screen **V10** shown in FIG. **15**. Specifically, in the region **V11**, a warning indicating a state anomaly such as a temperature anomaly or the like, information indicating a state in which some functions or manipulations are restricted, such as during execution of processing or during holding, or the like is displayed. In addition, in the region **V11**, information explicitly indicating that a setting different from the default state (for example, the standard state) has been made, such as a state in which some functions have been disabled (that is, have been turned off) or the like, may be displayed.

(139) In addition, FIG. **23** shows an example of the display information **V115** indicating a manipulation assigned to the up-down button **122** to be superimposed on an image displayed in the region **V11** in the screen **V10** shown in FIG. **15**. Examples of the manipulation assigned to the up-down button **122** include zoom control, mode switching, switching of an imaging device to be a control target in the multi mode, and the like. In addition, in the case where a manipulation via the up-down button **122** is temporarily restricted, information indicating that the manipulation is restricted may be displayed as the display information **V115**.

(140) A specific example of display information to be displayed in each of the regions **V11**, **V13**, and **V15** in the screen **V10** has been described above with reference to FIG. **15** and FIG. **19** to FIG. **23**.

(141) {1.4.4. UI of Imaging Device}

(142) Subsequently, an example of a UI of the imaging device **20** according to the present embodiment will be described with reference to FIG. **24**, particularly paying attention to a screen presented on the display unit **211** of the imaging device **20**. FIG. **24** is an explanatory diagram for describing an example of the UI of the imaging device **20** according to the present embodiment.

(143) As shown in FIG. **24**, a screen **V20** displayed on the display unit **211** includes a region **V21** and a region **V22**. The region **V22** is a region for displaying information indicating the settings and states related to image capturing performed by the imaging device **20**. In addition, the region **V21** is a region for displaying information indicating other states other than the settings and states related to image capturing performed by the imaging device **20**.

(144) For example, in the example shown in FIG. **24**, pieces of display information **V221** to **V228** are displayed in the region **V22**, and are equivalent to the display information **V151** to **V156**, **V112**,

**V115**, and **V116** on the screen **V10** (see FIG. 15) of the control device **10**, respectively. More specifically, the display information **V221** indicates the operation mode of the imaging device **20**, and is equivalent to the display information **V151** on the screen **V10**. In addition, the display information **V222** indicates the setting of a viewing angle, and is equivalent to the display information **V156** on the screen **V10**. In addition, the display information **V223** indicates the zoom magnification in image capturing, and is equivalent to the display information **V115** and **V116** on the screen **V10**. In addition, the display information **V224** indicates the format of video, and is equivalent to the display information **V153** on the screen **V10**. In addition, the display information **V225** indicates the mode of video (for example, the number of frames or the like), and is equivalent to the display information **V154** on the screen **V10**. In addition, the display information **V226** indicates the setting of image stabilization, and is equivalent to the display information **V155** on the screen **V10**. In addition, the display information **V227** indicates that the sound collection unit **270** of the imaging device **20** is off (that is, in a state in which sound is not collected), and is equivalent to the display information **V112** on the screen **V10**. In addition, the display information **V228** indicates the capturing time of video, and is equivalent to the display information **V152** on the screen **V10**.

(145) In addition, in the example shown in FIG. 24, pieces of display information **V211** to **V214** are displayed in the region **V21**, and are equivalent to the display information **V131** to **V133** and **V135** on the screen **V10** (see FIG. 15) of the control device **10**, respectively. More specifically, the display information **V211** indicates the state of communication with the control device **10** via a network based on the Wi-Fi standard, and is equivalent to the display information **V131** on the screen **V10**. In addition, the display information **V212** indicates the state of communication with the control device **10** via a network based on the Bluetooth standard, and is equivalent to the display information **V132** on the screen **V10**. In addition, the display information **V213** indicates the state of a positioning function such as GPS, and is equivalent to the display information **V133** on the screen **V10**. In addition, the display information **V214** indicates the remaining battery level of the imaging device **20**, and is equivalent to the display information **V135** on the screen **V10**. Note that, in the region **V21**, display information indicating a temperature anomaly of the imaging device **20** equivalent to the display information **V111** on the screen **V10** may be displayed.

(146) Note that, as shown in FIG. 24, as display information displayed on each of the regions **V21** and **V22** of the screen **V20**, display information with similar motifs to display information displayed on the screen **V10** is used. In addition, on the screen **V20**, similarly to the screen **V10** (see FIG. 15) of the control device **10**, the region **V22** in which information indicating the settings and states related to image capturing performed by the imaging device **20** is displayed is positioned on the lower side of the screen, and the region **V21** in which information indicating other states is displayed is positioned on the upper side of the screen. With such a configuration, it is possible for the user to intuitively recognize the position at which desired information is displayed also in the case of checking the screen **V20** of the imaging device **20** with a similar sense to the case of checking the screen **V10** of the control device **10**.

(147) In addition, the zoom magnification in image capturing presented as the display information **V223** is changed by a manipulation via the upper button **222** and the lower button **223** described with reference to FIG. 10. That is, the correspondence of the display information **V223** with the upper button **222** and the lower button **223** is similar to the correspondence between the display information **V115** and the up-down button **122** in the control device **10**. Thus, the display information **V223** is displayed in a region positioned close to the upper button **222** and the lower button **223** on the screen **V20** (that is, on the right end side of the screen **V20**), similarly to the display information **V115** on the screen **V10** of the control device **10**. Thus, it is possible for the user to intuitively manipulate the zoom magnification via the upper button **222** and the lower button **223** of the imaging device **20** with a similar sense to the case of manipulating the zoom magnification via the up-down button **122** of the control device **10**.

(148) Note that, in the case where the imaging device **20** is connected to the control device **10** via the network, regarding some manipulations, a manipulation via the control device **10** may only be enabled, and a manipulation via the input interface of the imaging device **20** may be restricted in order to prevent an incorrect manipulation. As a specific example, regarding the manipulation of the zoom magnification of the imaging device **20**, a manipulation via the up-down button **122** of the control device **10** may be enabled, and a manipulation via the upper button **222** and the lower button **223** of the imaging device **20** may be disabled.

(149) An example of the UI of the imaging device **20** according to the present embodiment has been described above with reference to FIG. **24**, particularly paying attention to the screen presented on the display unit **211** of the imaging device **20**. Note that the screen **V20** shown in FIG. **24** is a mere example, and the UI of the imaging device **20** is not necessarily limited only to the screen **V20**. As a specific example, a screen (that is, a screen including regions equivalent to the regions **V11**, **V13**, and **V15**) having a similar layout to the screen **V10** (see FIG. **15**) and the control device **10** may be presented as the UI of the imaging device **20**.

(150) {1.4.5. Configuration of Manipulation Menu}

(151) Subsequently, an example of manipulation menus of the control device **10** and the imaging device **20** according to the present embodiment will be described with reference to FIG. **25** and FIG. **26**. Here, the manipulation menus are UIs for selectively executing various functions (for example, changing of setting, execution of processing, and the like) of the control device **10** and the imaging device **20**.

(152) For example, FIG. **25** is an explanatory diagram for describing an example of a manipulation menu **V30** of the control device **10** according to the present embodiment. The manipulation menu **V30** shown in FIG. **25** is displayed on the display unit **111** in the case where the menu button **124** described with reference to FIG. **2** and FIG. **4** is pressed down, for example. Specifically, when the menu button **124** is pressed down, a screen **V31** is displayed first.

(153) In the manipulation menu **V30**, various functions of the control device **10** and the imaging device **20** are classified in advance in accordance with the type and target, and a manipulation screen for executing each of the functions is managed for each of the classes. For example, in the example shown in FIG. **25**, various functions of the control device **10** and the imaging device **20** are classified into “mode switching”, “setting related to image capturing”, “image reproduction”, “setting of imaging device”, “setting of control device”, and “communication between imaging device and control device”, and managed. Thus, on the screen **V31**, display information (for example, an icon) indicating the class of each of “mode switching”, “setting related to image capturing”, “image reproduction”, “setting of imaging device”, “setting of control device”, and “communication between imaging device and control device” is presented. By manipulating a predetermined button such as the up-down button **122** or the recording button **121**, for example, to select display information corresponding to a desired class among respective pieces of presented display information, the user can instruct the control device **10** to display a manipulation screen for the function assigned to the class. Note that the present description will be made assuming that display information associated with “setting related to image capturing” indicated by a reference character **V311** has been selected.

(154) When the display information **V311** is selected, the screens **V32** to **V34** for selecting respective functions classified as “setting related to image capturing” are displayed on the display unit **111**. On the screens **V32** to **V34**, display information indicating the respective functions classified as “setting related to image capturing” is displayed. For example, on the screen **V32**, display information indicating functions of checking and changing the setting of each of “resolution and number of frames”, “image format”, “image stabilization”, “viewing angle”, “zoom magnification”, and “turning of screen corresponding to orientation of imaging device” is presented. In addition, on the screen **V33**, display information indicating functions of checking and changing the setting of each of “exposure (brightness of captured image)”, “screen size”, “auto

white balance”, “screen effect”, “on/off of speaker”, and “noise cancelling function of speaker” is presented. By manipulating a predetermined button such as the up-down button **122** or the recording button **121** and switching the screen among the screens **V32** to **V34** to select display information associated with a desired function, the user can instruct the control device **10** to display a manipulation screen for the function.

(155) Note that, as display information displayed on the screens **V32** to **V34**, motifs similar to the display information to be displayed on the screen **V10** described with reference to FIG. **15** are used. Thus, it is possible for the user to intuitively recognize to which setting each piece of display information is related.

(156) With the configuration as described above, when the display information indicating the function of checking and changing the setting of “image stabilization” indicated by a reference character **V321** on the screen **V32** is selected, for example, a screen **V35** for checking and changing the setting of “image stabilization” is displayed. Then, by selecting display information associated with a desired set value from among pieces of display information associated with the respective set values of “image stabilization” presented on the screen **V35**, the setting of “image stabilization” is changed.

(157) In the above manner, on the manipulation menu **V30**, the respective screens are associated hierarchically in accordance with the class. With such a configuration, by selecting display information displayed on the screen as appropriate in accordance with an object to cause a manipulation screen corresponding to a desired function to be displayed, and performing a manipulation in accordance with the manipulation screen, it is possible for the user to cause the control device **10** to execute the function.

(158) Next, an example of a manipulation menu **V40** of the imaging device **20** according to the present embodiment will be described with reference to FIG. **26**. The manipulation menu **V40** shown in FIG. **26** is displayed on the display unit **211** in the case where the menu button **225** described with reference to FIG. **10** is pressed down, for example.

(159) Note that, as is understood when comparing FIG. **26** and FIG. **25**, the manipulation menu **V40** has a similar screen configuration to the manipulation menu **V30** of the control device **10** except that display information (for example, an icon or the like) is not displayed for a function not existing on the imaging device **20** side. That is, a screen **V41** in the manipulation menu **V40** is equivalent to the screen **V31** in the manipulation menu **V30**. Similarly, screens **V42** to **V45** in the manipulation menu **V40** are equivalent to the screens **V32** to **V35** in the manipulation menu **V30**. In addition, also for display information (an icon or the like) displayed on each of the screens **V41** to **V45** in the manipulation menu **V40**, similar motifs to the display information displayed on each of the screens **V31** to **V35** in the manipulation menu **V30** are used.

(160) With such a configuration, it is possible for the user to intuitively manipulate the manipulation menu **V40** of the imaging device **20** with a similar sense to the case of manipulating the manipulation menu **V30** of the control device **10**.

(161) An example of the manipulation menus of the control device **10** and the imaging device **20** according to the present embodiment has been described above with reference to FIG. **25** and FIG. **26**.

(162) <1.5. Evaluation>

(163) As described above, the screen **V10** presented as the UI of the control device **10** according to the present embodiment includes the region **V11** for presenting an image, and the regions **V13** and **V15** for presenting information corresponding to various states and settings, as shown in FIG. **15**. In the regions **V13** and **V15**, information set in advance as a presentation target in accordance with the operation state (operation mode) of the control device **10** is displayed constantly in a predetermined layout as long as the operation state is continued. On the other hand, various types of information superimposed on an image displayed in the region **V11** is temporarily displayed in accordance with a predetermined trigger, such as the case where a state in which image capturing is



difficult has been detected, for example, and is not displayed in the case where the imaging device **20** and the control device **10** are in the standard state.

(164) With such a configuration, for example, it is possible for the user to instruct the imaging device **20** to capture an image such as a moving image or a still image via the control device **10** while checking both of a preview image and various types of information (for example, setting, operation mode, state, and the like) related to image capturing. In addition, in the standard state, information is not superimposed on the image displayed in the region **V11**, and thus, the occurrence of such a situation in which a part of the image is shielded by the information is suppressed. That is, it is possible for the user to capture an image in a more suitable environment.

(165) In addition, in the case where a transition is made to a state different from the standard state, such as a temperature anomaly of the imaging device **20** or the like, information (for example, a warning or the like) indicating the state different from the standard state, which is hidden in the standard state, is superimposed on the image displayed in the region **V11**. With such a configuration, it is possible to raise user awareness that a transition is being made to a state different from the standard state (for example, an abnormal state is being sensed).

(166) In addition, among pieces of display information displayed on the screen **V10** of the control device **10**, display information associated in advance with the input interface such as a button provided for the control device **10** may be controlled in display position so as to be positioned close to the input interface. This also applies to the case of turning the housing **11** of the control device **10** such that the top and the bottom of the screen **V10** are reversed. With such a configuration, even if the user turns the screen **V10** with respect to the housing **11** such that it is easier to manipulate a predetermined input interface with his/her own dominant hand, display information associated with the input interface is displayed close to the input interface. Thus, it is possible for the user to perform a more intuitive manipulation corresponding to his/her usage form.

(167) In addition, in the present embodiment, display information with similar motifs to display information displayed on the screen **V10** (see FIG. 15) of the control device **10** is used as display information to be displayed on the screen **V20** (see FIG. 24) of the imaging device **20**. In addition, on the screen **V20**, the region **V22** in which information indicating the settings and states related to image capturing performed by the imaging device **20** is displayed is positioned on the lower side of the screen, and the region **V21** in which information indicating other states is displayed is positioned on the upper side of the screen, similarly to the screen **V10** of the control device **10**. With such a configuration, also in the case of checking the screen **V20** of the imaging device **20**, it is possible for the user to intuitively recognize the position at which desired information is displayed with a similar sense to the case of checking the screen **V10** of the control device **10**.

(168) In addition, in the present embodiment, the manipulation menu **V30** (FIG. 25) of the control device **10** and the manipulation menu **V40** (FIG. 26 and) of the imaging device **20** have a similar screen configuration, and display information with similar motifs is used as display information to be displayed on each of the screens. With such a configuration, it is possible for the user to perform an intuitive manipulation with a similar sense both in the case of manipulating the manipulation menu **V40** of the imaging device **20** and in the case of manipulating the manipulation menu **V30** of the control device **10**.

## 2. Second Embodiment

(169) Next, as a second embodiment of the present disclosure, an example of the UI of the control device **10** in the aforementioned multi mode (that is, a mode in which a plurality of imaging devices **20** are connected to the control device **10**) will be described.

### (170) <2.1. System Configuration>

(171) First, an example of a system configuration of an information processing system **2** according to the present embodiment will be described with reference to FIG. 27. FIG. 27 is an explanatory diagram for describing an example of a schematic system configuration of the information processing system **2** according to the present embodiment, and shows an example of a system

configuration during the operation in the multi mode.

(172) As shown in FIG. 27, in the information processing system 2 according to the present embodiment, the control device 10 and each of the plurality of imaging devices 20 are connected so as to be capable of communicating with each other via the wireless network N1. For example, in the example shown in FIG. 27, each of imaging devices 20a to 20b is connected to the control device 10 via the network N1. More specifically, in the case of applying the network N1 based on the Wi-Fi standard, it is possible to achieve a communication form as shown in FIG. 27 by causing the control device 10 to operate as an access point and connecting each of the imaging devices 20a to 20c as a station to the control device 10.

(173) Note that communication between the control device 10 and the imaging devices 20a to 20c via the network N1 can be connected and disconnected individually on the basis of a standard such as Wi-Fi or the like, for example. Thus, for example, it is also possible to dynamically increase or decrease the number of the imaging devices 20 to be connected to the control device 10.

(174) On the basis of such a configuration, the control device 10 is configured to be capable of controlling the operation (for example, the operation related to image capturing) of each of the imaging devices 20a to 20c connected via the network N1. At this time, the control device 10 may individually control the operation of each of the imaging devices 20a to 20c via the network N1. In addition, the control device 10 may collectively control the operation of each of the imaging devices 20a to 20c via the network N1.

(175) In addition, the control device 10 may present an image (for example, a preview image) captured by at least any imaging device 20 among the imaging devices 20a to 20c and transmitted from the imaging device 20 via the network N1 to the user via the display unit 111. In addition, the control device 10 may acquire information indicating the setting and state of each of the imaging devices 20a to 20c from each of the imaging devices 20a to 20c via the network N1, and may present the acquired information to the user via the display unit 111.

(176) An example of the schematic system configuration of the information processing system according to the second embodiment of the present disclosure has been described above with reference to FIG. 27.

(177) <2.2. Technical Characteristics>

(178) Next, as technical characteristics of the information processing system 2 according to the present embodiment, an example of a UI presented by the control device 10 during the operation in the multi mode will be particularly described with reference to FIG. 28. FIG. 28 is an explanatory diagram for describing an example of the UI of the control device 10 according to the present embodiment, and shows an example of a screen displayed as a UI on the display unit 111 of the control device 10. Note that, in the present description, it is assumed that each of the imaging devices 20a to 20e is connected to the control device 10 via the network N1.

(179) As shown in FIG. 28, a screen V50 displayed on the display unit 111 includes a region V51 for presenting an image (for example, a preview image), and regions V53 and V55 for presenting information corresponding to various states, settings, and the like. For example, in the example shown in FIG. 28, the strip region V53 with the lateral direction serving as the longitudinal direction is provided above the region V51. Similarly, the strip region V55 with the lateral direction serving as the longitudinal direction is provided under the region V51. Such a configuration is similar to the configuration (that is, the positional relationship among the regions V11, V13, and V15) of the screen V10 of the control device 10 according to the first embodiment described with reference to FIG. 15.

(180) In the region V53, information indicating other states other than the settings and states related to image capturing is displayed. Note that the configuration of the region V53 is similar to the configuration of the region V13 in the screen V10 of the control device 10 according to the first embodiment except that information on the imaging device 20 side is not presented. For example, in the example shown in FIG. 28, pieces of display information V531 to V534 are displayed in the

region V53. Here, the display information V531, V532, V533, and V534 in FIG. 28 correspond to V131, V132, V134, and V136 in the screen V10 shown in FIG. 15.

(181) On the other hand, on the screen V50 according to the present embodiment, information V561 indicating the settings and states related to image capturing of each of the plurality of imaging devices 20 connected to the control device 10 is displayed in the region V55 in association with regions different from one another in the region V55 for each of the imaging device 20. For example, in the example shown in FIG. 28, in each of regions denoted by the reference characters V56a to V56e in the region V55, pieces of the information V561 indicating an operation mode and an imaging state (for example, a recording state) are displayed for the imaging devices 20 different from one another. For example, in the region V56a, the information V561 indicating the operation mode and imaging state of the imaging device 20a is displayed. Similarly, in the regions V56b to V56e, pieces of the information V561 indicating the operation modes and imaging states of the imaging devices 20b to 20e are displayed.

(182) Note that the order of displaying the respective pieces of information V561 of the plurality of imaging devices 20 in the region V53 may be controlled in accordance with a predetermined condition. As a specific example, the respective pieces of information V561 of the plurality of imaging devices 20 may be displayed in accordance with the order of connection to the control device 10 via the network N1. In addition, as another example, the respective pieces of information V561 of the plurality of imaging devices 20 may be displayed in the order corresponding to identification information (for example, number or the like) set in advance for each of the plurality of imaging devices 20. In addition, as another example, the respective pieces of information V561 of the plurality of imaging devices 20 may be displayed in the order corresponding to the positional relationship of the respective plurality of imaging devices 20. More specifically, control may be exerted such that the information V561 of the imaging device 20 positioned closer to the control device 10 is displayed by higher priority (for example, to be positioned closer to the left side). In addition, as another example, the respective pieces of information V561 of the plurality of imaging devices 20 may be displayed in the order corresponding to the respective states of the plurality of imaging devices 20. As a more specific example, control may be exerted such that the information V561 of the imaging device 20 having a higher remaining battery level among the plurality of imaging devices 20 is displayed by higher priority (for example, to be positioned closer to the left side). As a matter of course, the example described above is a mere example, and the method is not particularly limited as long as weights can be assigned to the plurality of imaging devices 20 in accordance with a predetermined condition, and the order of displaying the information V561 of each of the plurality of imaging devices 20 can be controlled in accordance with the weights.

(183) In addition, information indicating the setting of the order of displaying the respective pieces of information V561 of the plurality of imaging devices 20 in the region V53 may be held in a predetermined storage area. With such a configuration, when the plurality of imaging devices 20 are connected to the control device 10, for example, it is also possible to cause the respective pieces of information V561 of the plurality of imaging devices 20 to be displayed in the region V53 in a similar order to a previous connection.

(184) In the region V11, a so-called preview image, an actually captured image, and the like are displayed. In addition, in the region V11, information indicating various settings and states is superimposed on a displayed image in some cases.

(185) For example, in the example shown in FIG. 28, pieces of display information V511 to V514 corresponding to various settings and states of the control device 10 are displayed in the region V11. Here, the display information V511, V513, and V514 in FIG. 28 correspond to V113, V114, and V115 on the screen V10 shown in FIG. 15. In addition, the display information V512 indicates that the control device 10 is operating in the multi mode. Note that pieces of the display information V511 to V514 are temporarily displayed in accordance with a predetermined trigger, and are not displayed in the case where the control device 10 is in the standard state.

(186) In addition, in the region **V51**, display information **V562** corresponding to the state of each of the plurality of imaging devices **20** is displayed for each of the imaging devices **20** in some cases. At this time, for example, the display information **V562** corresponding to the state of the imaging device **20a** is displayed close to the information **V561** of the imaging device **20a** displayed in the region **V56a** in the region **V55**. For example, in the example shown in FIG. **28**, the display information **V562** corresponding to the state of the imaging device **20a** is displayed at a position equivalent to directly above the display information **V561** of the imaging device **20a** displayed in the region **V56a** in a region close to the lower end of the region **V51**. In addition, for each of the imaging devices **20b** to **20e**, the display information **V562** of each of the imaging devices **20** is similarly displayed close to the display information **V561** of the imaging device **20** displayed in the region **V55**.

(187) In addition, the display information **V562** of each of the respective imaging devices **20** is displayed temporarily in accordance with a predetermined trigger, and is not displayed in the case where the imaging device **20** is in the standard state. As a specific example, in the case where a state anomaly such as a temperature anomaly or the like or a state in which some functions or manipulations are restricted, such as during execution of processing, is detected in each of the imaging devices **20**, display information corresponding to the state is displayed as the display information **V562** of the imaging device **20**.

(188) Note that the manner of displaying an image such as a preview image in the region **V51** is not particularly limited. For example, only images captured by some of the imaging devices **20** which are selected by the user from the plurality of imaging devices **20** connected to the control device **10** may be displayed in the region **V51**. In addition, as another example, by dividing the region **V51** into a plurality of regions, an image captured by each of the plurality of imaging devices **20** may be caused to be displayed in each of the divided regions.

(189) In addition, in the state where each of the imaging devices **20** is capturing an image in the multi mode, at least some of manipulations via the input interface of each of the control device **10** and the imaging device **20** may be restricted in order to prevent an incorrect manipulation. As a specific example, in the state where each of the imaging devices **20** is capturing an image in the multi mode, the manipulation of the zoom magnification of each of the imaging devices **20** may be restricted. In this case, control may be exerted such that, in the state where each of the imaging devices **20** is not capturing an image, for example, the manipulation of the zoom magnification of each of the imaging devices **20** is enabled. As a matter of course, regardless of the state of each of the imaging devices **20**, a manipulation via the input interface of each of the control device **10** and the imaging device **20** may be enabled at all the times. In addition, at least some of the plurality of imaging devices **20** may be configured such that a manipulation on each of the imaging devices **20** can be set as presetting.

(190) As technical characteristics of the information processing system **2** according to the present embodiment, an example of the UI presented by the control device **10** during the operation in the multi mode has been particularly described above with reference to FIG. **28**.

(191) <2.3. Evaluation>

(192) As described above, the screen **V50** presented as the UI of the control device **10** according to the present embodiment includes the region **V51** for presenting an image and the regions **V53** and **V55** for presenting information corresponding to various states, settings, and the like, as shown in FIG. **28**. In the regions **V53** and **V55**, information set in advance as a presentation target in accordance with the operation state (operation mode) of the control device **10** is displayed constantly in a predetermined layout as long as the operation state is continued. Note that, on the other hand, various types of information superimposed on an image displayed in the region **V11** is temporarily displayed in accordance with a predetermined trigger such as the case where a state in which image capturing is difficult is detected or the like, for example, and is not displayed in the case where the imaging device **20** and the control device **10** itself are in the standard state.

(193) In particular, on the screen **V50**, the display information **V561** corresponding to the state of each of the plurality of imaging devices **20** connected to the control device **10** is displayed constantly in the region **V55**. With such a configuration, it is possible for the user to always grasp the state of each of the imaging devices **20**. In addition, information is not superimposed on an image displayed in the region **V51** in the standard state, and thus, the occurrence of such a situation in which a part of the image is shielded by the information is suppressed. That is, it is possible for the user to capture an image in a more suitable environment.

(194) In addition, in the case where at least any of the plurality of imaging devices **20** transitions to a state different from the standard state, such as a temperature anomaly, information (for example, a warning or the like) indicating the state, which is hidden in the standard state, is superimposed on the image displayed in the region **V51**. With such a configuration, it is possible to raise user awareness that a transition is being made to a state different from the standard state (for example, an abnormal state is being sensed). In addition, at this time, the display information **V562** of each of the plurality of imaging devices **20** is displayed to be positioned close to the display information **V561** of the imaging device **20** displayed in the region **V55**. With such a configuration, it is possible for the user to more intuitively grasp the imaging device **20** transitioned to a state different from the standard state among the plurality of imaging devices **20** connected to the control device **10**.

### 3. Hardware Configuration Example

(195) Next, a hardware configuration of the control device **10** according to an embodiment of the present disclosure will be described with reference to FIG. **29**. FIG. **29** is a block diagram illustrating a hardware configuration example of the control device **10** according to an embodiment of the present disclosure.

(196) As illustrated in FIG. **29**, the control device **10** includes a central processing unit (CPU) **901**, a read only memory (ROM) **903**, and a random access memory (RAM) **905**. In addition, the control device **10** may include a host bus **907**, a bridge **909**, an external bus **911**, an interface **913**, an input device **915**, an output device **917**, a storage device **919**, a drive **921**, a connection port **923**, and a communication device **925**. Further, the control device **10** may include an imaging device **933** and a sensor **935** as necessary. The control device **10** may include a processing circuit such as a digital signal processor (DSP) or an application specific integrated circuit (ASIC) instead of or in addition to the CPU **901**.

(197) The CPU **901** serves as an operation processing device and a control device, and controls all of or a part of the operations in the control device **10** in accordance with various programs recorded in the ROM **903**, the RAM **905**, the storage device **919** or a removable recording medium **927**. The ROM **903** stores programs and operation parameters used by the CPU **901**. The RAM **905** temporarily stores program used in the execution of the CPU **901** and parameters that changes as appropriate in the execution. The CPU **901**, ROM **903**, and RAM **905** are connected to each other by the host bus **907** including an internal bus such as a CPU bus. In addition, the host bus **907** is connected to the external bus **911** such as a peripheral component interconnect/interface (PCI) bus via the bridge **909**. Note that, each of the configuration of the above-described control unit **140** may be implemented by the CPU **904**, for example.

(198) The input device **915** is, for example, a device manipulated by a user with a mouse, a keyboard, a touch panel, buttons, switches, a lever, and the like. The input device **915** may include a mic that detects a sound of a user. The input device **915** may be, for example, a remote control unit using infrared light or other radio waves, or may be an external connection device **929** such as a portable phone operable in response to the manipulation of the control device **10**. The input device **915** includes an input control circuit that generates an input signal on the basis of the information input by a user and outputs the input signal to the CPU **901**. By manipulating the input device **915**, a user can input various types of data to the control device **10** or issue instructions for causing the control device **10** to perform a processing operation. In addition, the imaging device

**933** to be described below can function as an input device by imaging a motion or the like of a hand of the user. Note that, the above-described input unit **120** may be implemented by the input device **915**, for example.

(199) The output device **917** includes a device capable of visually or audibly notifying a user of the acquired information. The output device **917** may be, for example, a display device such as a liquid crystal display (LCD), a plasma display panel (PDP), an organic electro-luminescence (EL) display, and a projector, a hologram display device, an audio output device, such as a speaker or a headphone, and a printer. The output device **917** outputs the results obtained from the process of the control device **10** in a form of video such as text or an image and voice or sound. In addition, the output device **917** may include a light or the like to brighten the surroundings. Note that, the above-described output unit **110** may be implemented by the output device **917**, for example.

(200) The storage device **919** is a device for data storage configured as an example of a storage unit of the control device **10**. The storage device **919** includes, for example, a magnetic storage device such as a hard disk drive (HDD), a semiconductor storage device, an optical storage device, or a magneto-optical storage device. The storage device **919** stores programs to be executed by the CPU **901**, various data, and data obtained from the outside. Note that the above-described storage unit **150** may be implemented by the storage device **919**, for example.

(201) The drive **921** is a reader/writer for the removable recording medium **927** such as a magnetic disk, an optical disk, a magneto-optical disk, or a semiconductor memory, and is embedded in the control device **10** or externally attached thereto. The drive **921** reads information recorded in the attached removable recording medium **927** and outputs the information to the RAM **905**. Further, the drive **921** writes record in the attached removable recording medium **927**.

(202) The connection port **923** is a port used to directly connect devices to the control device **10**. The connection port **923** may be, for example, a universal serial bus (USB) port, an IEEE1394 port, a small computer system interface (SCSI) port, or the like. In addition, the connection port **923** may be an RS-232C port, an optical audio terminal, High-Definition Multimedia Interface (HDMI) (registered trademark) port, or the like. By connecting the external connection device **929** to the connection port **923**, various data may be exchanged between the control device **10** and the external connection device **929**.

(203) The communication device **925** is, for example, a communication interface including a communication device or the like for connection to a communication network **931**. The communication device **925** may be, for example, a communication card for a wired or wireless local area network (LAN), Bluetooth (registered trademark), wireless USB (WUSB), or the like. In addition, the communication device **925** may be a router for optical communication, a router for asymmetric digital subscriber line (ADSL), a modem for various kinds of communications, or the like. The communication device **925** transmits and receives signals to and from, for example, the Internet or other communication devices using a predetermined protocol such as TCP/IP. In addition, the communication network **931** to be connected to the communication device **925** is a network connected in a wired or wireless manner, and is, for example, the Internet, a home LAN, infrared communication, radio wave communication, satellite communication, or the like. Note that the above-described communication unit **130** may be implemented by the communication device **925**, for example.

(204) The imaging device **933** is a device that generates an image by imaging a real space using an image sensor such as a charge-coupled device (CCD) or a complementary metal oxide semiconductor (CMOS), as well as various members such as one or more lenses for controlling the formation of a subject image on the image sensor, for example. The imaging device **933** may be a device that captures still images and may also be a device that captures moving images. Note that the above-described imaging unit **207** may be implemented by the imaging device **933**, for example.

(205) The sensor **935** is any of various sensors such as an acceleration sensor, a gyro sensor, a

geomagnetic sensor, an optical sensor, or a sound sensor, for example. The sensor 935 acquires information regarding the state of the control device 10, such as the attitude of the case of the control device 10, as well as information regarding the environment surrounding the control device 10, such as brightness or noise surrounding the control device 10, for example. The sensor 935 may also include a global positioning system (GPS) sensor that receives GPS signals and measures the latitude, longitude, and altitude of the device.

(206) The foregoing thus illustrates an exemplary hardware configuration of the control device 10. Each of the above components may be realized with general-purpose members or hardware specialized in the function of each component. Such a configuration may also be modified as appropriate in accordance with the technological level at the time of the implementation.

(207) Further, it is also possible to create a program for causing hardware such as a processor, a memory, and a storage incorporated into a computer to exert a function equivalent to the structural elements included in the above-described control device 10. In addition, it is also possible to provide a computer readable storage medium in which the program is recorded.

#### 4. Conclusion

(208) The preferred embodiment(s) of the present disclosure has/have been described above with reference to the accompanying drawings, whilst the present disclosure is not limited to the above examples. A person skilled in the art may find various alterations and modifications within the scope of the appended claims, and it should be understood that they will naturally come under the technical scope of the present disclosure.

(209) Further, the effects described in this specification are merely illustrative or exemplified effects, and are not limitative. That is, with or in the place of the above effects, the technology according to the present disclosure may achieve other effects that are clear to those skilled in the art from the description of this specification.

(210) Additionally, the present technology may also be configured as below.

(211) (1)

(212) An information processing apparatus including: a display control unit configured to cause an image captured by any of one or a plurality of imaging devices to be displayed in a captured image display region that is a part of a display unit, cause second information related to the one or plurality of imaging devices to be displayed in a captured image hiding region that is a part of the display unit and different from the captured image display region, and cause first information related to the one or plurality of corresponding imaging devices to be displayed in the captured image display region in accordance with the second information being displayed and the one or plurality of imaging devices being in a predetermined state.

(2)

(213) The information processing apparatus according to (1), in which the second information is information related to an operation state of the one or plurality of imaging devices and displayed constantly.

(3)

(214) The information processing apparatus according to (1) or (2), in which the first information is information related to capturing of the image or restriction of a manipulation performed by the one or plurality of imaging devices.

(4)

(215) The information processing apparatus according to any one of (1) to (3), in which an operation state of the imaging devices is a state related to a state or setting related to capturing of the image performed by the one or plurality of imaging devices.

(5)

(216) The information processing apparatus according to any one of (1) to (4), including: a communication unit configured to communicate with the imaging devices, in which the display control unit causes the image captured by the imaging devices to be displayed in the captured

image display region.

(6)

(217) The information processing apparatus according to (5), including: an imaging control unit configured to, on a basis of communication with the imaging devices via the communication unit, control an operation of the imaging devices.

(7)

(218) The information processing apparatus according to any one of (1) to (4), including: at least one of the one or plurality of imaging devices, in which the display control unit causes the image captured by the imaging device to be displayed in the captured image display region.

(8)

(219) The information processing apparatus according to any one of (1) to (7), in which the display control unit causes the second information related to the plurality of imaging devices to be displayed, and causes first information related to the plurality of imaging devices corresponding to the second information to be displayed in accordance with the second information being displayed and a state of the plurality of imaging devices.

(9)

(220) The information processing apparatus according to (8), in which the display control unit causes the first information to be displayed in the captured image display region for each of the plurality of imaging devices in association with the second information associated with the imaging device, in accordance with the state of each of the imaging devices.

(10)

(221) The information processing apparatus according to (8) or (9), in which the display control unit causes the second information associated with each of the plurality of imaging devices to be displayed in the captured image hiding region in an order corresponding to a predetermined condition.

(11)

(222) The information processing apparatus according to any one of (1) to (10), in which the display control unit presents a plurality of the captured image hiding regions on the display unit, and causes the plurality of respective captured image hiding regions to display pieces of information different from one another as the second information.

(12)

(223) The information processing apparatus according to (11), in which the display control unit causes information corresponding to a state or setting related to capturing of the image to be displayed in some of the plurality of captured image hiding regions.

(13)

(224) The information processing apparatus according to (11), in which the display control unit causes information corresponding to another state different from the state related to capturing of the image to be displayed in some of the plurality of captured image hiding regions.

(14)

(225) The information processing apparatus according to any one of (1) to (13), including: a manipulation unit configured to accept a manipulation on the display unit in a predetermined direction; and a housing configured to hold the display unit and the manipulation unit such that the display unit and the manipulation unit are positioned close to each other, in which the display control unit causes display information corresponding to a manipulating direction of the manipulation unit to be displayed as the first information in a portion of the captured image display region positioned close to the manipulation unit, in accordance with a display state of information on the display unit.

(15)

(226) The information processing apparatus according to (14), in which the display control unit causes the display information to be displayed in the captured image display region such that the



manipulating direction of the manipulation unit and a direction indicated by the display information displayed in the captured image display region substantially match.

(16)

(227) An information processing method including, by a processor: causing an image captured by any of one or a plurality of imaging devices to be displayed in a captured image display region that is a part of a display unit; causing second information related to the one or plurality of imaging devices to be displayed in a captured image hiding region that is a part of the display unit and different from the captured image display region; and causing first information related to the one or plurality of corresponding imaging devices to be displayed in the captured image display region in accordance with the second information being displayed and the one or plurality of imaging devices being in a predetermined state.

(17)

(228) A program causing a computer to execute: causing an image captured by any of one or a plurality of imaging devices to be displayed in a captured image display region that is a part of a display unit; causing second information related to the one or plurality of imaging devices to be displayed in a captured image hiding region that is a part of the display unit and different from the captured image display region; and causing first information related to the one or plurality of corresponding imaging devices to be displayed in the captured image display region in accordance with the second information being displayed and the one or plurality of imaging devices being in a predetermined state.

(18)

(229) An information processing system including: one or a plurality of imaging devices configured to capture an image; and a display control device including a communication unit configured to communicate with the one or plurality of imaging devices and a display control unit configured to cause the image captured by any of the one or plurality of imaging devices to be displayed in a captured image display region that is a part of a display unit, in which the display control unit causes second information related to the one or plurality of imaging devices to be displayed in a captured image hiding region that is a part of the display unit and different from the captured image display region, and causes first information related to the one or plurality of corresponding imaging devices to be displayed in the captured image display region in accordance with the second information being displayed and the one or plurality of imaging devices being in a predetermined state.

(19)

(230) The information processing system according to (18), including: the plurality of imaging devices, in which in accordance with a state or setting of each of the plurality of imaging devices, the display control unit causes the second information to be displayed in the captured image hiding region for each of the imaging devices in association with the imaging devices.

#### REFERENCE SIGNS LIST

(231) **1, 2** information processing system **10** control device **11** housing **110** output unit **111** display unit **120** input unit **121** recording button **122** up-down button **123** power button **124** menu button **130** communication unit **140** control unit **141** input analysis unit **143** output control unit **150** storage unit **191** projecting portion **192** connector **193** fitting portion **20** imaging device **21** housing **22** lens unit **207** imaging unit **210** output unit **211** display unit **220** input unit **221** recording button **222** upper button **223** lower button **224** power button **225** menu button **230** communication unit **240** control unit **241** imaging control unit **243** input analysis unit **245** output control unit **250** storage unit **260** imaging unit **261** lens optical system **270** sound collection unit **291** fitting portion **292** memory card slot **293** battery cover **294** lock switch

#### Claims

1. An information processing system, comprising: a first processor configured to: control capture of an image; and control detection of a temperature anomaly of an information processing apparatus; and a second processor configured to: control display of a captured image in a first region, wherein the first region is a part of a display region of a display device; control display of first information in a second region, wherein the first information is related to an imaging operation associated with the captured image, the second region is a part of the display region of the display device, and the second region is positioned at one of a top of the display device or a bottom of the display device; and control superimposed display of a warning of a detected temperature anomaly on the captured image, wherein the warning of the detected temperature anomaly is displayed in the first region based on the detection of the temperature anomaly, and the warning of the detected temperature anomaly is concurrently displayed with the first information.
  2. The information processing system according to claim 1, wherein the detection of the temperature anomaly is based on a value of a temperature associated with the information processing apparatus, and the temperature is outside of a specific temperature range.
  3. The information processing system according to claim 1, wherein the first information is associated with an operation state of the information processing apparatus, and the first information is displayed constantly.
  4. The information processing system according to claim 3, wherein the operation state of the information processing apparatus is one of a state or a setting related to the captured image.
  5. The information processing system according to claim 1, wherein the second processor is further configured to superimpose display of second information over the captured image, in the first region.
  6. The information processing system according to claim 5, wherein the second information includes an indication of one of execution of a processing operation or a hold operation.
  7. The information processing system according to claim 5, wherein the second processor is further configured to control the superimposed display of the second information on the captured image based on a transition of an operation state of the information processing apparatus from a standard state to a non-standard state, and an operation of the information processing apparatus has an anomaly in the non-standard state.
  8. The information processing system according to claim 7, wherein manipulation of the display device via an input button is temporarily restricted in the non-standard state.
  9. The information processing system according to claim 8, wherein the manipulation of the display device is to change brightness of the captured image.
  10. The information processing system according to claim 1, wherein the second processor is further configured to control display of second information in the second region.
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